foot size and are also manufactured specifically to cater to the occasional oddity. Slyvia Fado creates high-end designer heels and uses 3D printers to make hydraulic heels with the ZMorph multitool 3D printer. The shapes and patterns of 3D printed shoes are mostly unique, geometric and can sport a very different look-and-feel.

12 MOVIE PROPS

Another surprising application of 3D printing is the creation of props used in films. This technology can build props quickly and at an affordable price. Therefore, a few film sets have already begun keeping a 3D printer handy to quickly print any prop they might require while shooting, specifically catered to the film's aesthetics and set design. Costumes, especially for superheroes and sci-fi characters, can also be 3D printed. Black Panther (2018) used 3D printing to create extravagant accessories for Queen Ramonda, T'Challa's mother. Jurassic World (2015) also utilised 3D printers to make realistic-looking dinosaur skeletons.

13 MUSICAL INSTRUMENTS

Many musicians are using 3D printing to bring their artistic creations of musical instruments to life while still being affordable. You can print any musical instrument, from guitars to clarinets, with "print your own custom instrument" plans. The 3D Varius is a high-end electric, 3D printed violin which is actually based on a Stradivarius model. The instrument is lightweight with an intricate design and has a ton of functionalities that normal violins do not possess. 3D printed violins and fiddles have been used by professional musicians in orchestras.

Functional violins are being created using 3D printing technology

14 SPORTS EQUIPMENT

You can now 3D print sports equipment like running shoes, golf clubs, mouthguards, shin guards and more. Reebok has even come out with a 3D printed sensor that evaluates head injuries and flashes one of three LED indicators depending on the seriousness of the injury. Other examples of 3D printing sports equipment include Proto3000 and Thingiverse bikes. Proto3000 prints surfboards that are extremely lightweight and functional. Thingiverse offers plans that enable customers to 3D print their own bikes with replaceable parts. If a part of the bike breaks, you can just replace it by printing a new one.

15 3D PRINTING PENS

3D printing pens are used to create 3D drawings by drawing in the air. 3Doodler, the 3D printing pen, uses quick-hardening filament as ink which solidifies as the user is drawing in the air. You can draw free-form, three-dimensional objects with your hand which eliminates having to make concept designs on the computer first. This pen is very useful for professionals like architects, artists, interior designers, and other designers. Users can control the speed and amount of ink (filament) that comes out of the pen to make precise and accurate drawings of their projects.

16 MARS COLONY

3D printing has traversed all the way to the red planet now. Researchers in Illinois are developing a 3D printing method that can create tools and structures out of Martian dust. Yes, you heard that right. The dirt on Mars can actually be transformed into a rubber-like material which can be used for making small things like tools or huge structures and buildings as well. This will save explorers a lot of time and money since they will not have to ship building materials to Mars from Earth. The process can help save considerable cost for a Mars mission, in terms of the fuel needed to ship all the extra weight to Mars. An entire colony can be built from scratch by people who come to this planet in the future using 3D printers to convert Martian dirt to buildings and structures. The martian soil is known as regolith, and is a great raw material for 3D printing.

Simulated Martian dust was used to create these tools

17 ROBOTS

Robots are being built using 3D printing as well. At MIT, a small-sized robot was built using this technology. Apart from the motor, everything that made up the robot's body was 3D printed in one piece. There's very little assembly involved in building the robot and absolutely no materials are wasted. 3D printed robots are quite inexpensive. This paves a way for mass production of robots, owing to their affordability, for usage in homes, factories and offices.

18 CASTS

A Russian company has begun manufacturing medical casts using 3D printing. These modern-day casts have quite a few advantages over traditional plaster-based casts. Firstly, they are water-friendly, therefore you can go about your routine activities like bathing or showering without having to worry about it. Secondly, the casts have a geometric design, therefore, there are plenty of holes present in them, so you can easily scratch an itch. Also, you can avoid the nasty rashes and infections that you can get from traditional casts.

Apart from the ones we have mentioned in our lists, there are several other bizarre and outlandish applications of 3D printing and new ones are coming out every day. 3D printing's biggest use is still surrounds prototyping and innovation, however, the creation of products and commodities are also becoming a common practice.